

Subdomain enumeration

Subdomain enumeration is the process of finding valid subdomains for a domain, but why do we do this? We do this to expand our attack surface to try and discover more potential points of vulnerability.

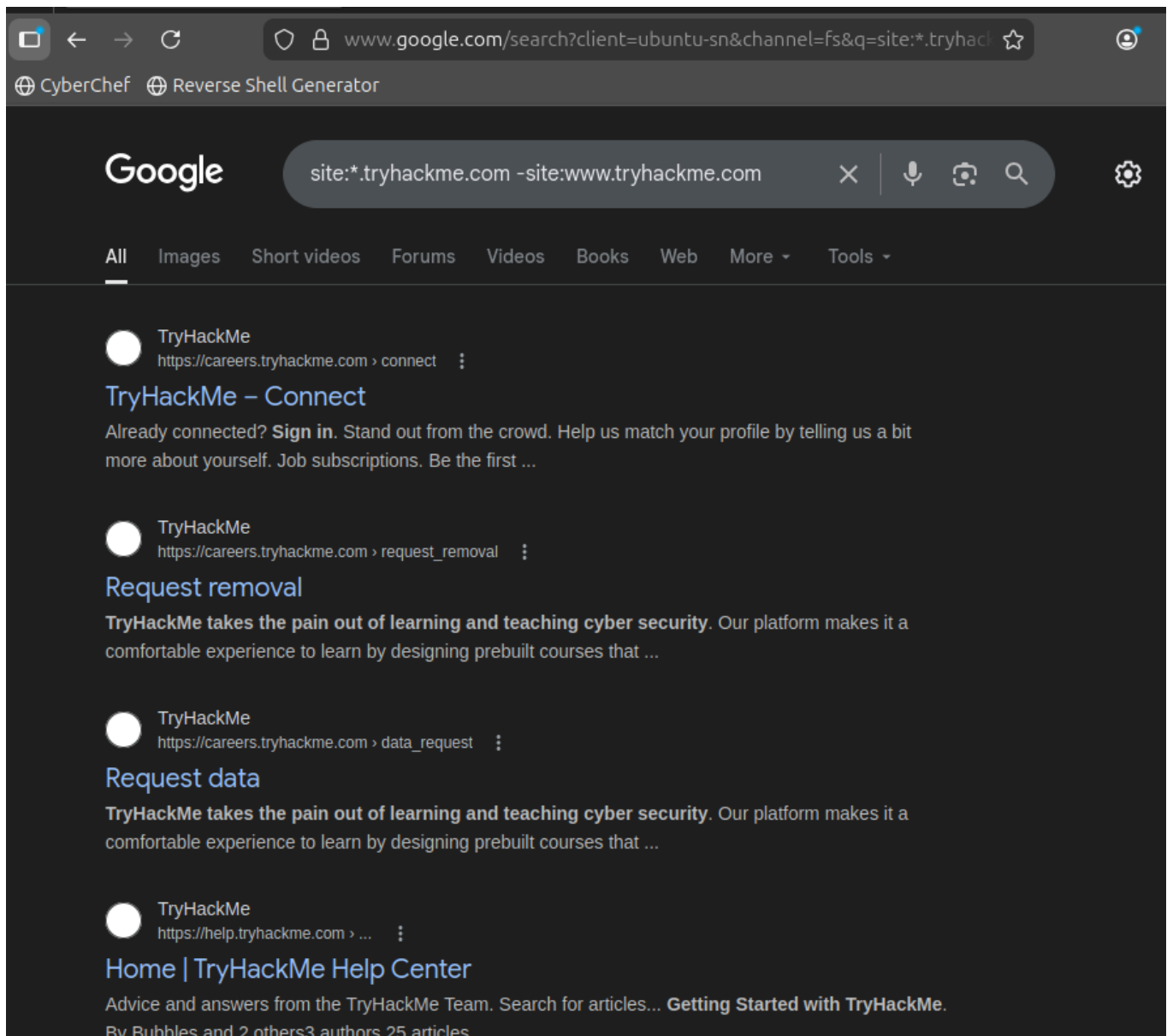
OSINT - SSL/TLS Certificates

When an SSL/TLS (Secure Sockets Layer/Transport Layer Security) certificate is created for a domain by a CA (Certificate Authority), CA's take part in what's called "Certificate Transparency (CT) logs".

We can use this service to our advantage to discover subdomains belonging to a domain, sites like <https://crt.sh>[\(opens in new tab\)](#) offer a searchable database of certificates that shows current and historical results.

OSINT - Search Engines

Search engines contain trillions of links to more than a billion websites, which can be an excellent resource for finding new subdomains



DNS Bruteforce

Bruteforce DNS (Domain Name System) enumeration is the method of trying tens, hundreds, thousands or even millions of different possible subdomains from a pre-defined list of commonly used subdomains.

```
user@thm:~$ dnsrecon -t brt -d acmeitsupport.thm
[*] No file was specified with domains to check.
[*] Using file provided with tool: /usr/share/dnsrecon/namelist.txt
[*] A api.acmeitsupport.thm 10.10.10.10
[*] A www.acmeitsupport.thm 10.10.10.10
[+] 2 Record Found
user@thm:~$
```

OSINT - Sublist3r

To speed up the process of OSINT subdomain discovery, we can automate the above methods with the help of tools like [Sublist3r](#)

```
user@thm:~$ ./sublist3r.py -d acmeitsupport.thm

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# Coded By Ahmed Aboul-Ela - @aboul3la

[-] Enumerating subdomains now for acmeitsupport.thm
[-] Searching now in Baidu..
[-] Searching now in Yahoo..
[-] Searching now in Google..
[-] Searching now in Bing..
[-] Searching now in Ask..
[-] Searching now in Netcraft..
[-] Searching now in Virustotal..
[-] Searching now in ThreatCrowd..
[-] Searching now in SSL Certificates..
[-] Searching now in PassiveDNS..
[-] Searching now in PassiveDNS..
[-] Searching now in Virustotal..
[-] Total Unique Subdomains Found: 2
web55.acmeitsupport.thm
www.acmeitsupport.thm
user@thm:~$
```

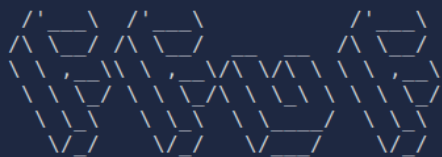
Virtual Hosts

Some subdomains aren't always hosted in publically accessible DNS results, such as development versions of a web application or administration portals

DNS record could be kept on a private DNS server or recorded on the developer's machines in their `/etc/hosts` file (or `c:\windows\system32\drivers\etc\hosts` file for Windows users), which

maps domain names to IP addresses.

```
root@ip-10-128-109-50:~# ffuf -w /usr/share/wordlists/SecLists/Discovery/DNS/namelist.txt -H "Host: FUZZ.acmeitsupport.thm" -u http://10.128.134.178 -fs 2395
```



v2.1.0-dev

```
:: Method      : GET
:: URL         : http://10.128.134.178
:: Wordlist    : FUZZ: /usr/share/wordlists/SecLists/Discovery/DNS/namelist.txt
:: Header     : Host: FUZZ.acmeitsupport.thm
:: Follow redirects : false
:: Calibration : false
:: Timeout     : 10
:: Threads    : 40
:: Matcher    : Response status: 200-299,301,302,307,401,403,405,500
:: Filter     : Response size: 2395
```

```
delta      [Status: 200, Size: 51, Words: 7, Lines: 1, Duration: 0ms]
yellow     [Status: 200, Size: 56, Words: 8, Lines: 1, Duration: 0ms]
:: Progress: [1907/1907] :: Job [1/1] :: 1250 req/sec :: Duration: [0:00:01] :: Errors: 0 ::
root@ip-10-128-109-50:~#
```